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Hydrocarbons

NEET Chemistry · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	105 questions
Answer key	at the end

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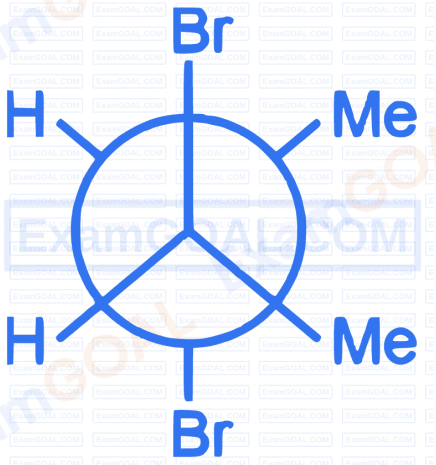
MCQ (Single Correct Answer) 105 questions

Q1. RE-NEET 2026

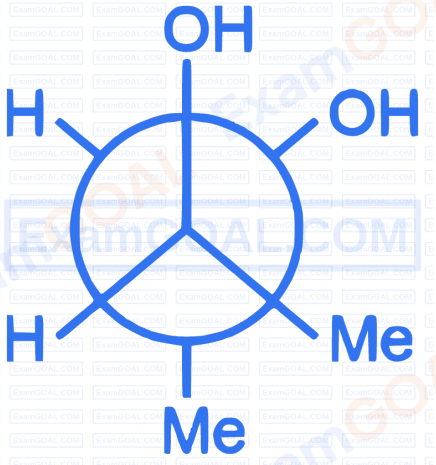
Open solution

Given below are two statements :

Statement I : trans-But-2-ene upon treatment with Br_2 in CCl_4 gives the following product.



Statement II : cis-But-2-ene upon treatment with alkaline $KMnO_4$ gives the following product.



In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Statement I is incorrect but Statement II is correct

(B) Both Statement I and Statement II are correct

(C) Both Statement I and Statement II are incorrect

(D) Statement I is correct but Statement II is incorrect

Q2. RE-NEET 2026

Open solution

Among the following, the compound having conjugated double bonds is

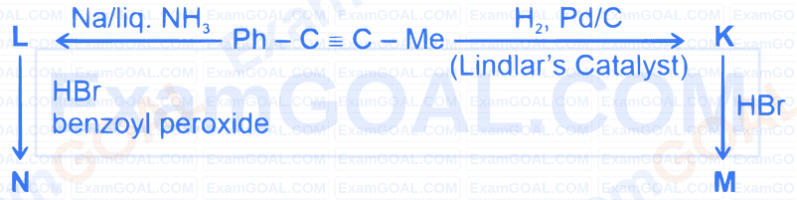
(A) hepta-1,6-diene (B) hepta-1,3-diene (C) hepta-1,4-diene (D) hepta-1,5-diene

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Q3. RE-NEET 2026

Open solution

Consider the following reaction sequences and choose the correct option.



(A) $\mathbf{M}$ and $\mathbf{N}$ are stereoisomers

(B) $\mathbf{K}$ and $\mathbf{L}$ are geometrical isomers

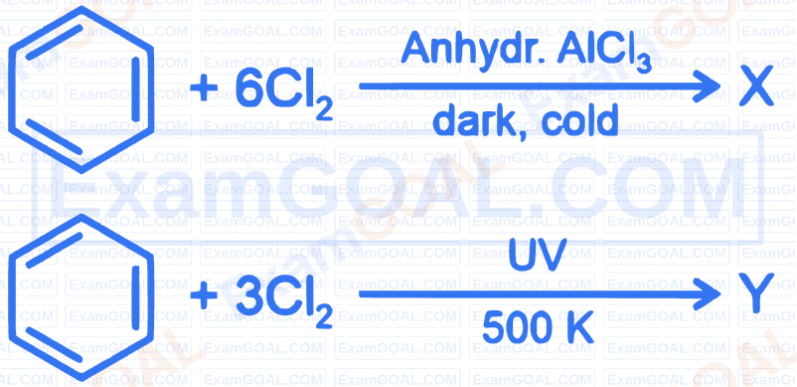
(C) $\mathbf{K}$ and $\mathbf{L}$ are enantiomers

(D) $\mathbf{M}$ and $\mathbf{N}$ are geometrical isomers

Q4. NEET 2026

Open solution

The number of chlorine atoms present in the organic products X and Y of the following reactions, respectively, are :



(A) 3 and 3

(B) 6 and 3

(C) 6 and 6

(D) 3 and 6

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Q5. NEET 2025

Which one of the following reactions does NOT give benzene as the product?

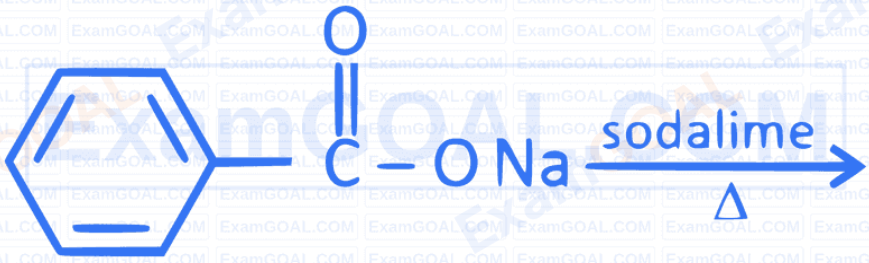
Open solution

(A) $\mathrm{H-C\equiv C-H} \xrightarrow[\text{at } 873\ \mathrm{K}]{\text{red hot Iron Tube}}$

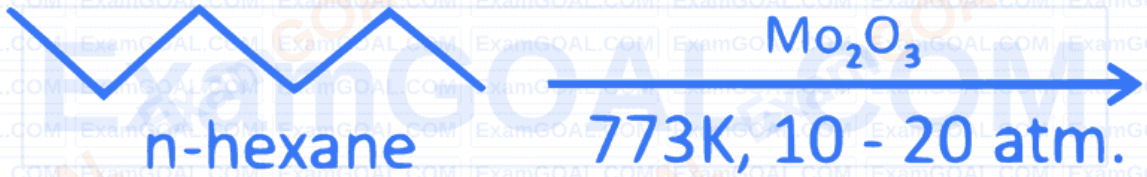
(B)



(C)



(D)



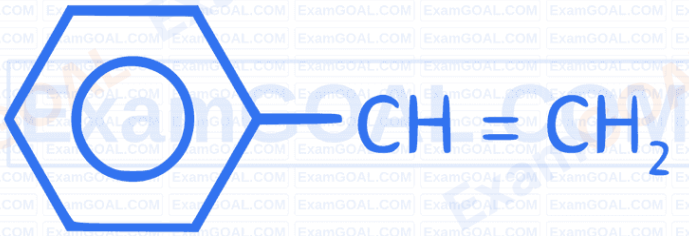
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Q6. NEET 2025

Which one of the following compounds does not decolourize bromine water?

Open solution

(A)



(B)



(C)



(D)

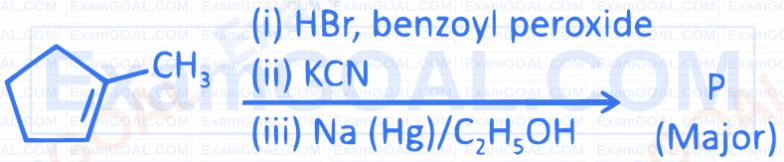


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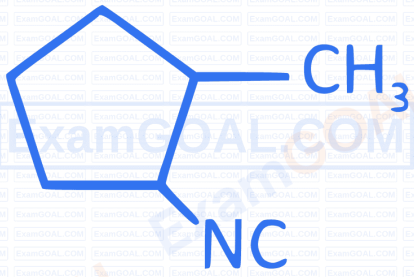
Q7. NEET 2025

Open solution

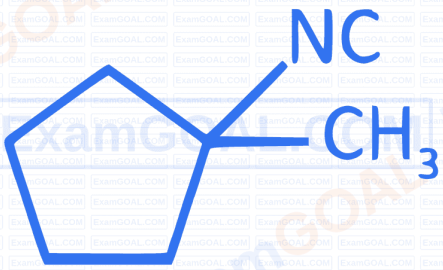
Predict the major product ' P ' in the following sequence of reactions-



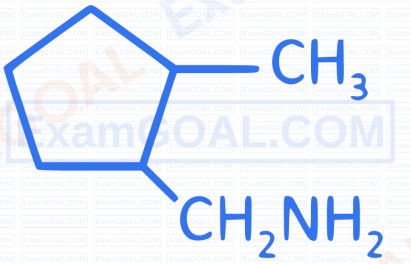
(A)



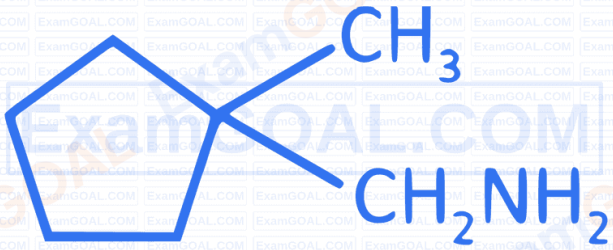
(B)



(C)



(D)



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Q8. NEET 2024 (Re-Examination)

Open solution

Given below are two statements:

Statement I: Propene on treatment with diborane gives an addition product with the formula

$$((CH_3)_2-CH\rightarrow)_3B$$

Statement II: Oxidation of

$$((CH_3)_2-CH\rightarrow)_3B$$

with hydrogen peroxide in presence of $NaOH$ gives propan-2-ol.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are correct

(D) Both Statement I and Statement II are incorrect

Q9. NEET 2024 (Re-Examination)

Open solution

The alkane that can be oxidized to the corresponding alcohol by $KMnO_4$ as per the equation



is, when :

(A) $R_1 = H; R_2 = H; R_3 = H$

(B) $R_1 = CH_3; R_2 = CH_3; R_3 = CH_3$

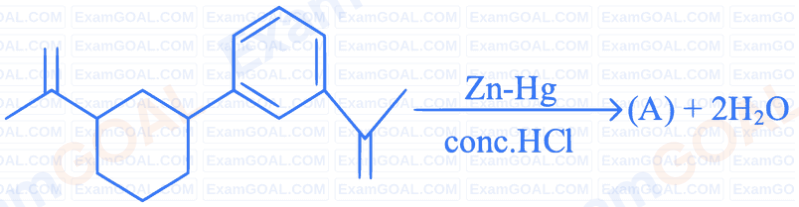
(C) $R_1 = CH_3; R_2 = H; R_3 = H$

(D) $R_1 = CH_3; R_2 = CH_3; R_3 = H$

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Q10. NEET 2023

Identify product (A) in the following reaction :



Open solution

(A)



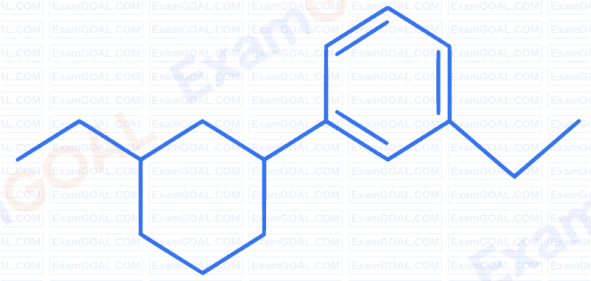
(B)



(C)



(D)



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Q11. NEET 2022 Phase 2

Open solution

The decreasing order of boiling points of the following alkanes is :

(a) heptane

(b) butane

(c) 2-methylbutane

(d) 2-methylpropane

(e) hexane

Choose the correct answer from the options given below :

(A) (a) > (e) > (c) > (b) > (d)

(B) (a) > (c) > (e) > (d) > (b)

(C) (c) > (d) > (a) > (e) > (b)

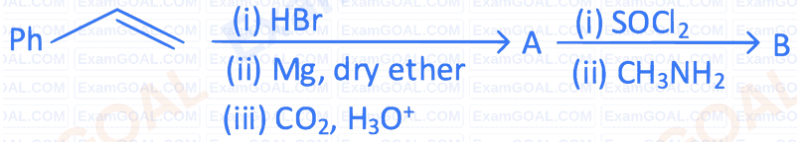
(D) (a) > (e) > (b) > (c) > (d)

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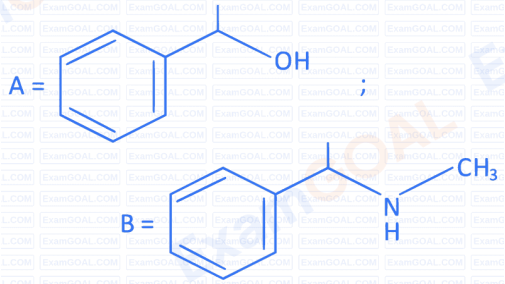
Q12. NEET 2022 Phase 2

Open solution

The products A and B in the following reaction sequence are :



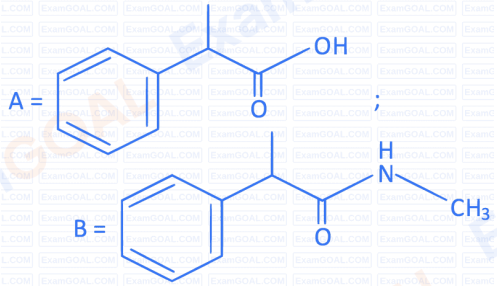
(A)



(B)



(C)



(D)



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Q13. NEET 2022 Phase 2

The incorrect method for the synthesis of alkenes is

(A) Treating vicinal dihalides with Zn metal

(B) Treating of alkynes with Na in liquid NH_3

(C) Heating alkyl halides with alcoholic KOH

(D) Treating alkyl halides in aqueous KOH solution

Open solution

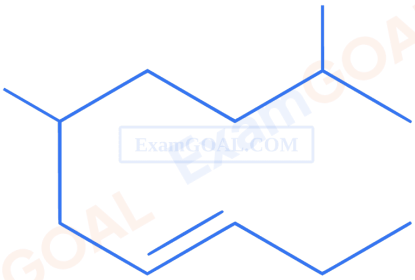
Q14. NEET 2021

The correct structure of 2, 6-dimethyl-dec-4-ene is

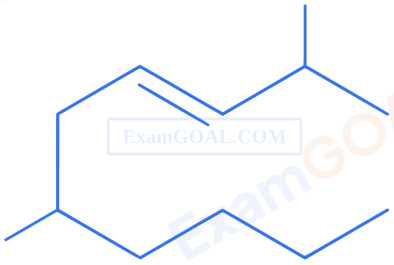
(A)



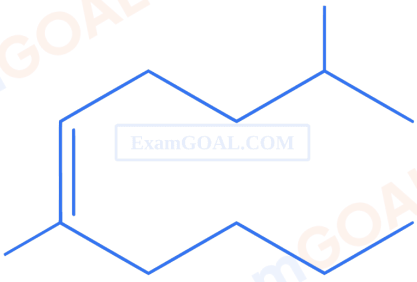
(B)



(C)



(D)



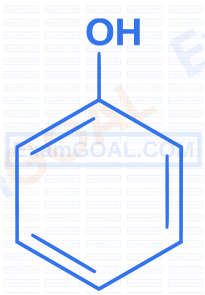
Open solution

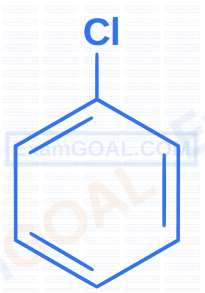
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Q15. NEET 2020 Phase 2

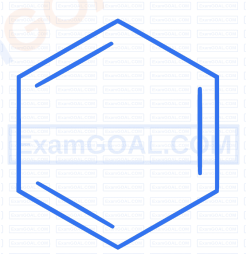
Open solution

Which of the following compound is most reactive in electrophilic aromatic substitution?

(A) 

(B) 

(C) 

(D) 

Q16. NEET 2020 Phase 2

Open solution

Which of the following is a free radical substitution reaction?

(A) Propene with $\mathrm{HBr}/(\mathrm{C_6H_5COO})_2$

(B) Benzene with $\mathrm{Br_2}/\mathrm{AlCl_3}$

(C) Acetylene with HBr

(D) Methane with $\mathrm{Br_2}/h\nu$

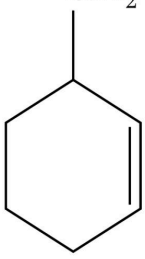
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Q17. NEET 2020 Phase 1

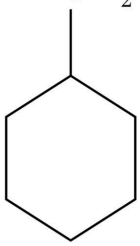
Open solution

An alkene on ozonolysis gives methanal as one of the products, Its structure is :

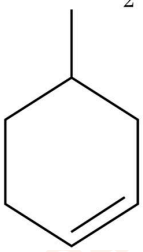
(A) $CH_2–CH_2–CH_3$



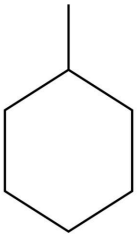
(B) $CH_2–CH=CH_2$



(C) $CH_2CH_2CH_3$



(D) $CH=CH_2–CH_3$



Q18. NEET 2019 (Odisha)

Open solution

The alkane that gives only one mono-chloro product on chlorination with Cl_2 in presence of diffused sunlight is

(A) Isopentane

(B) 2, 2-dimethylbutane

(C) neopentane

(D) n-pentane

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Q19. NEET 2019 (Odisha)

Open solution

In the following reaction,

$$H_3C-C\equiv CH \xrightarrow[873\ K]{\text{red hot iron tube}} A$$

the number of sigma (σ) bonds present in the product A, is

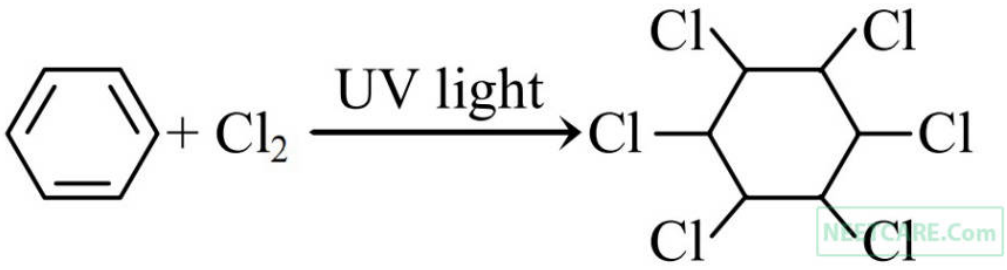
(A) 18 (B) 21 (C) 9 (D) 24

Q20. NEET 2019

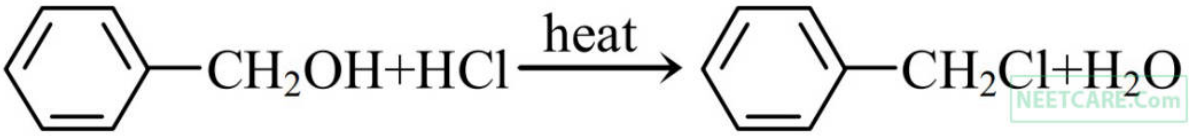
Open solution

Among the following, the reaction that proceeds through an electrophilic substitution, is :

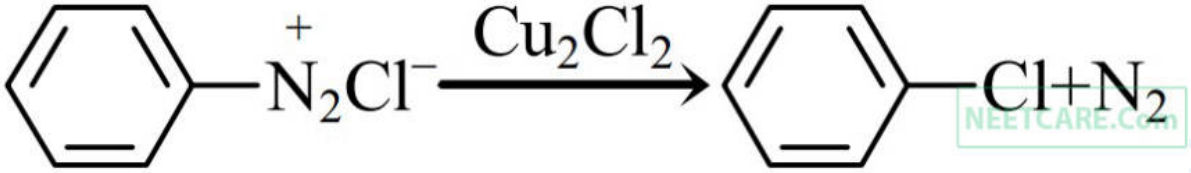
(A)



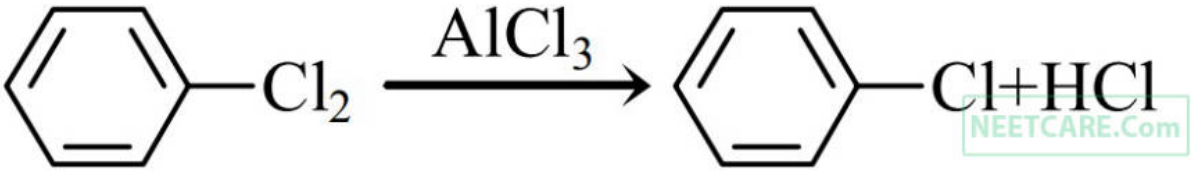
(B)



(C)



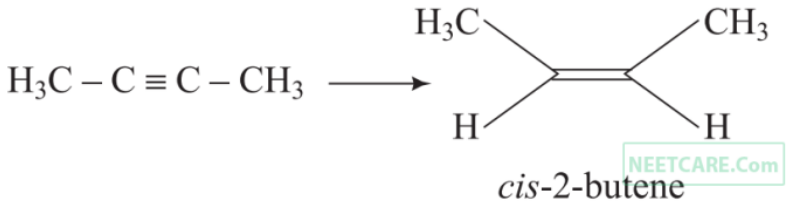
(D)



Q21. NEET 2019

Open solution

The most suitable reagent for the following conversion, is :



cis-2-butene

(A) Zn / HCl

(B) Hg^{2+} / H^+, H_2O

(C) Na / liquid NH_3

(D) H_2, Pd/C, quinoline

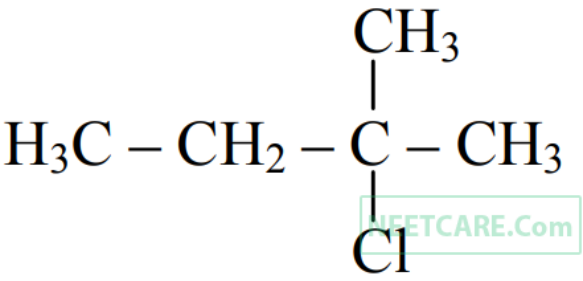
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Q22. NEET 2019

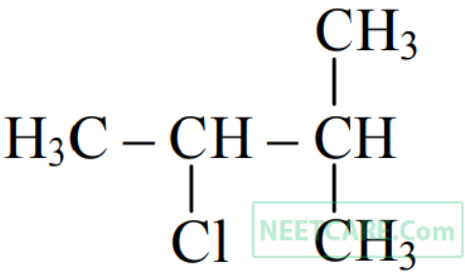
An alkene "A" on reaction with O_3 and Zn + H_2O gives propanone and ethanal in equimolar ratio. Addition of HCl to alkene "A" gives "B" as the major product. The structure of product "B" is :

Open solution

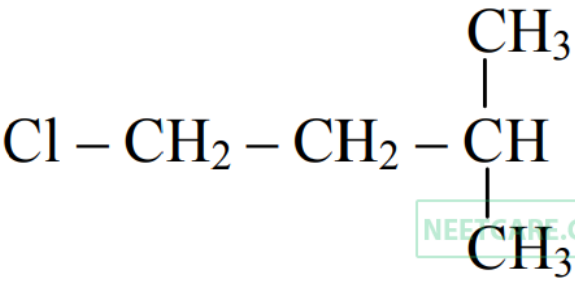
(A)



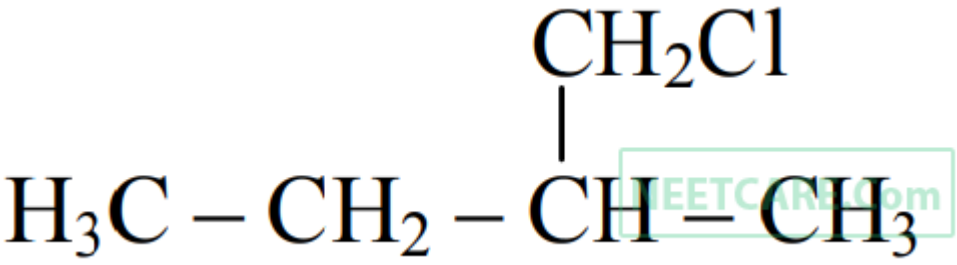
(B)



(C)



(D)



Q23. NEET 2018

Hydrocarbon (A) reacts with bromine by substitution to form an alkyl bromide which by Wurtz reaction is converted to gaseous hydrocarbon containing less than four carbon atoms. (A) is

Open solution

(A) $CH \equiv CH$ (B) $CH_2 = CH_2$ (C) $CH_3 - CH_3$ (D) CH_4

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Q24. NEET 2017

Open solution

Which one is the correct order of acidity?

(A) $CH \equiv CH > CH_3 - C \equiv CH$
$> CH_2 = CH_2 > CH_3 - CH_3$

(B) $CH \equiv CH > CH_2 = CH_2$
$> CH_3 - C \equiv CH > CH_3 - CH_3$

(C) $CH_3 - CH_3 > CH_2 = CH_2$
$> CH_3 - C \equiv CH > CH \equiv CH$

(D) $CH_2 = CH_2 > CH_3 - CH = CH_2$
$> CH_3 - C \equiv CH > CH \equiv CH$

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Q25. NEET 2017

Predict the correct intermediate and product in the following reaction :

$$H_3C-C\equiv CH \xrightarrow[HgSO_4]{H_2O,H_2SO_4} \underset{(A)}{Intermediate} \rightarrow \underset{(B)}{\text{Product}}$$

Open solution

(A)



(B)



(C)



(D)



Q26. NEET 2017

With respect to the conformers of ethane, which of the following statements is true?

(A) Bond angle changes but bond length remains same.

(B) Both bond angle and bond length change.

(C) Both bond angle and bond length remain same.

(D) Bond angle remains same but bond length changes.

Open solution

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Q27. NEET 2016 Phase 2

Which of the following can be used as the halide component for Friedel-Crafts reaction?

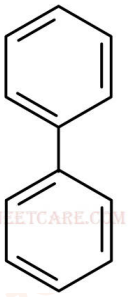
(A) Chlorobenzene
(B) Bromobenzene
(C) Chloroethene
(D) Isopropyl chloride

Open solution

Q28. NEET 2016 Phase 2

In which of the following molecules, all atoms are coplanar?

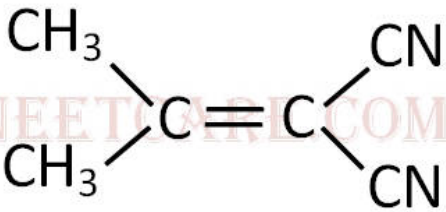
(A)



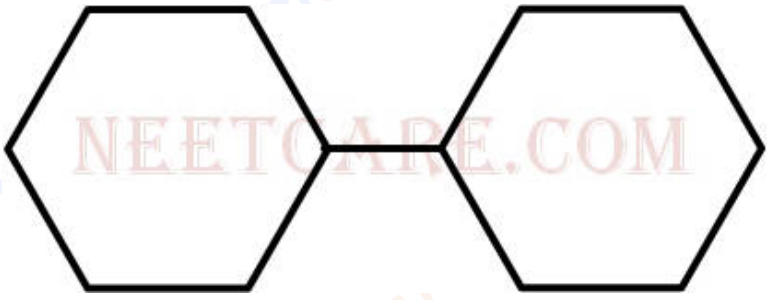
(B)



(C)



(D)

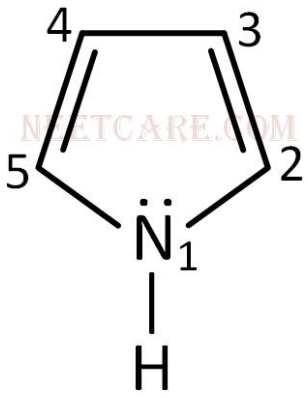


Open solution

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Q29. NEET 2016 Phase 2

In pyrrole the electron density is maximum on



Open solution

(A) 2 and 3 (B) 3 and 4 (C) 2 and 4 (D) 2 and 5

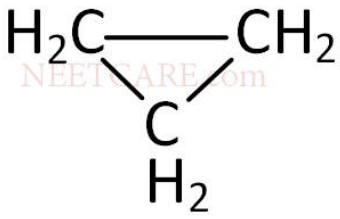
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Q30. NEET 2016 Phase 2

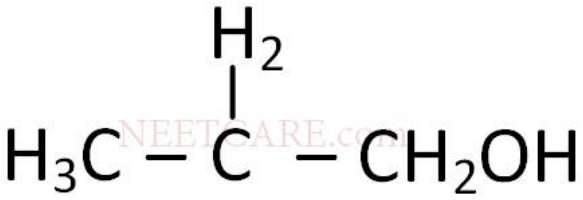
Which of the following compounds shall not produce propene by reaction with HBr followed by elimination or direct only elimination reaction ?

Open solution

(A)



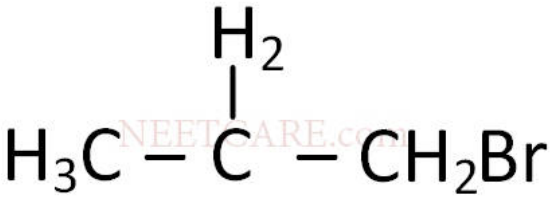
(B)



(C)



(D)



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Q31. NEET 2016 Phase 2

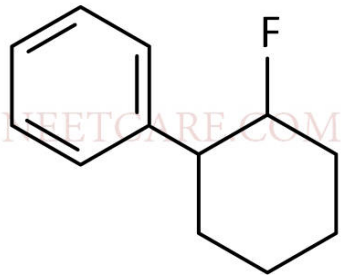
Open solution

In the given reaction,

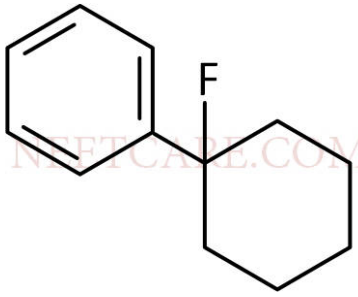


the product P is

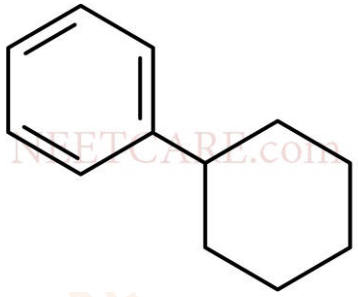
(A)



(B)



(C)



(D)



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Q32. NEET 2016 Phase 2

The compound that will react most readily with gaseous bromine has the formula

(A) C_3H_6 **(B)** C_2H_2 **(C)** C_4H_{10} **(D)** C_2H_4

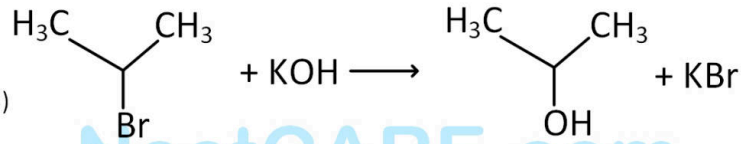
Open solution

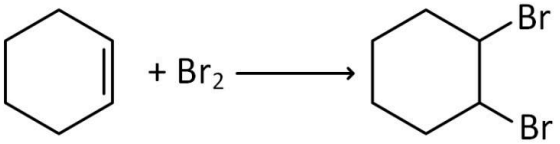
Q33. NEET 2016 Phase 1

For the following reactions :

(A) $CH_3CH_2CH_2Br + KOH \rightarrow$

$CH_3CH = CH_2 + KBr + H_2O$

(B) 

(C) 

NeetCARE.com

Which of the following statements is correct?

(A) (A) is elimination, (B) and (C) are substitution reactions.

(B) (A) is substitution, (B) and (C) are addition reactions.

(C) (A) and (B) are elimination reactions and (C) is addition reaction.

(D) (A) is elimination, (B) is substitution and (C) is addition reaction.

Open solution

Q34. NEET 2016 Phase 1

Consider the nitration of benzene using mixed conc.H_2SO_4 and HNO_3. If a large amount of $KHSO_4$ is added to the mixture, the rate of nitration will be

(A) unchanged **(B)** doubled **(C)** faster **(D)** slower

Open solution

Q35. NEET 2016 Phase 1

The pair of electrons in the given carbanion, $CH_3C \equiv C^-$, is present in which of the following orbitals?

(A) sp^2 **(B)** sp **(C)** 2p **(D)** sp^3

Open solution

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Q36. NEET 2016 Phase 1

Open solution

In the reaction

$$H-C\equiv CH \xrightarrow[(ii)\ CH_3CH_2Br]{(i)\ NaNH_2/liq.\ NH_3} X \xrightarrow[(ii)\ CH_3CH_2Br]{(i)\ NaNH_2/liq.\ NH_3} Y,$$

X and Y are

(A) X = 2-Butyne, Y = 2-Hexyne

(B) X = 1-Butyne, Y = 2-Hexyne

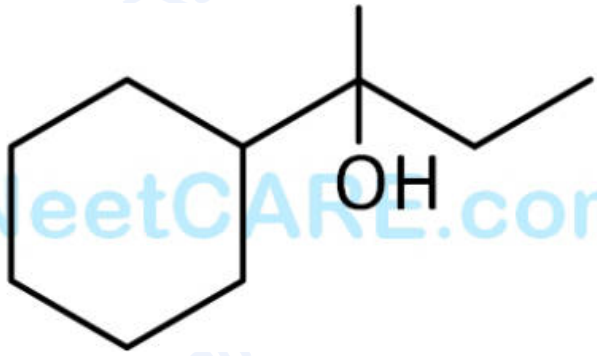
(C) X = 1-Butyne, Y = 3-Hexyne

(D) X = 2-Butyne, Y = 3-Hexyne

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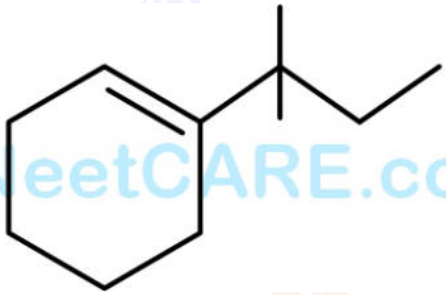
Q37. AIPMT 2015

Which of the following is not the product of dehydration of

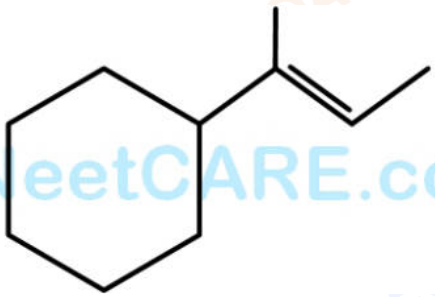


Open solution

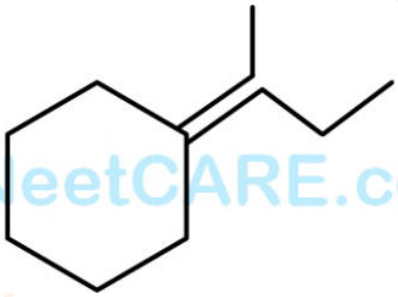
(A)



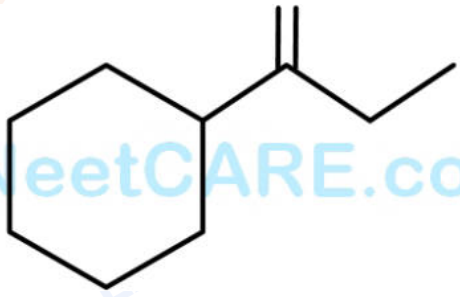
(B)



(C)



(D)



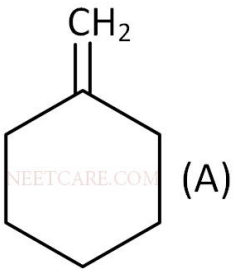
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Q38. AIPMT 2015

In the reaction with HCl, an alkene reacts in accordance with the Markovnikov's rule to give a product 1-chloro-1-methylcyclohexane. The possible alkene is

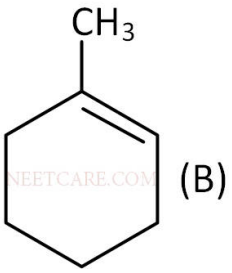
Open solution

(A)



(A)

(B)



(B)

(C) (A) and (B)

(D)



Q39. AIPMT 2015

2,3-Dimethyl-2-butene can be prepared by heating which of the following compounds with a strong acid?

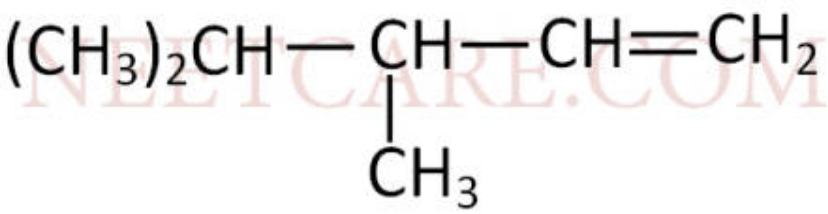
Open solution

(A) $(CH_3)_3CCH = CH_2$

(B) $(CH_3)_2C = CHCH_2CH_3$

(C) $(CH_3)_2CHCH_2CH = CH_2$

(D)

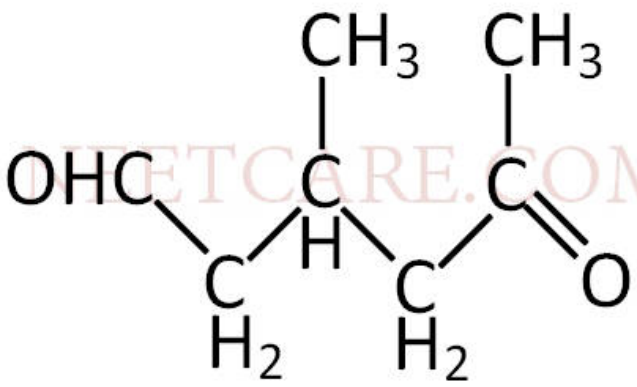


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Q40. AIPMT 2015 Cancelled Paper

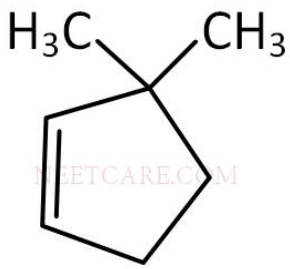
Open solution

A single compound of the structure,

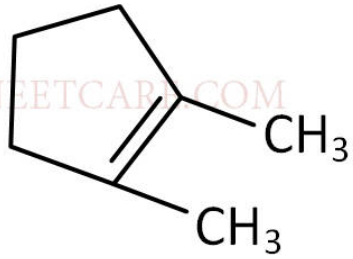


is obtainable from ozonolysis of which of the following cyclic compounds?

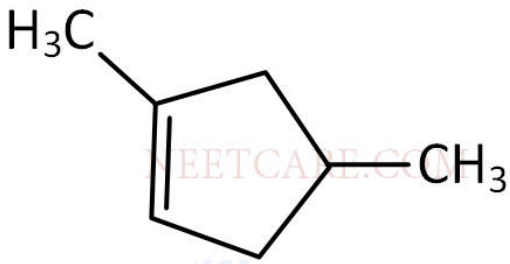
(A)



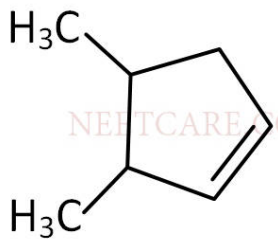
(B)



(C)



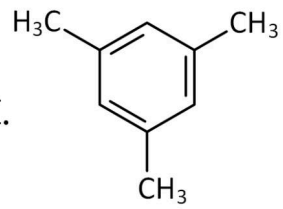
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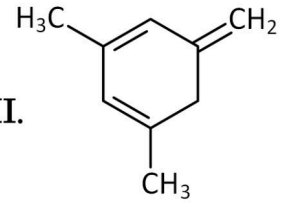
(D) 

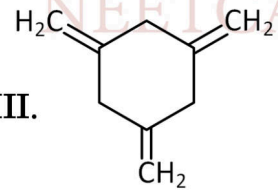
Q41. AIPMT 2015 Cancelled Paper

Open solution

Given :

I. 

II. 

III. 

The enthalpy of hydrogenation of these compounds will be in the order as

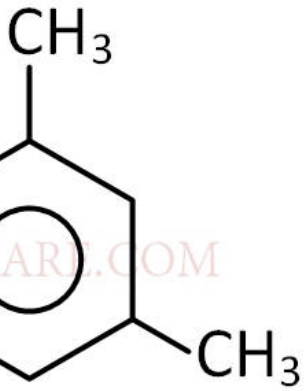
(A) II > III > I (B) II > I > III (C) I > II > III (D) III > II > I

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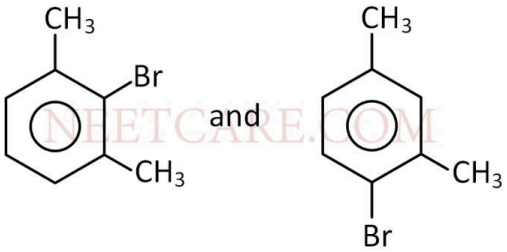
Q42. AIPMT 2014

What products are formed when the following compound is treated with Br_2 in the presence of $FeBr_3$?

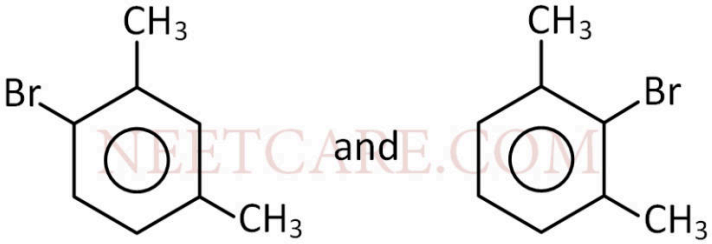
Open solution



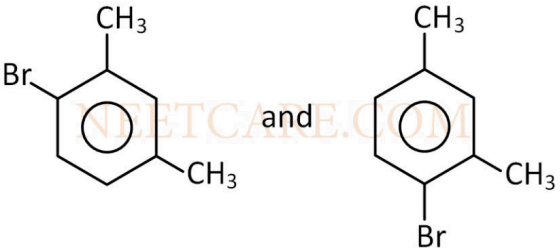
(A)



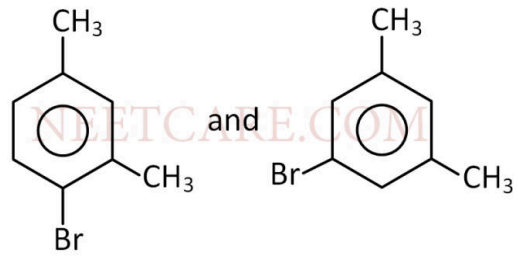
(B)



(C)

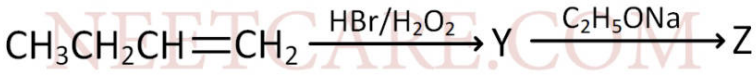


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(D) 

Q43. AIPMT 2014

Identify Z in the sequence of reactions :



Open solution

(A) $CH_3-(CH_2)_3-O-CH_2CH_3$

(B) $(CH_3)_2CH-O-CH_2CH_3$

(C) $CH_3(CH_2)_4-O-CH_3$

(D) $CH_3CH_2-CH(CH_3)-O-CH_2CH_3$

Q44. AIPMT 2014

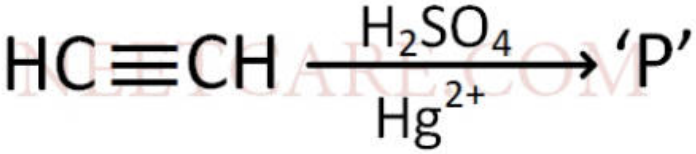
Which of the following organic compounds has same hybridization as its combustion product (CO_2)?

(A) Ethane (B) Ethyne (C) Ethene (D) Ethanol

Open solution

Q45. NEET 2013 (Karnataka)

In the following reaction :



Open solution

Product 'P' will not give

(A) Tollen's reagent test

(B) Brady's reagent test

(C) Victor Meyer test

(D) Iodoform test

Q46. NEET 2013

Which of the following compounds will not undergo Friedal-Craft's reaction easily?

(A) Nitrobenzene (B) Toluene (C) Cumene (D) Xylene

Open solution

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Q47. AIPMT 2012 Mains

Which of the following reagents will be able to distinguish between 1-butyne and 2-butyne?

(A) $NaNH_2$ (B) HCl (C) O_2 (D) Br_2

Open solution

Q48. AIPMT 2012 Prelims

Among the following compounds the one that is most reactive towards electrophilic nitration is

(A) benzoic acid (B) nitrobenzene (C) toluene (D) benzene

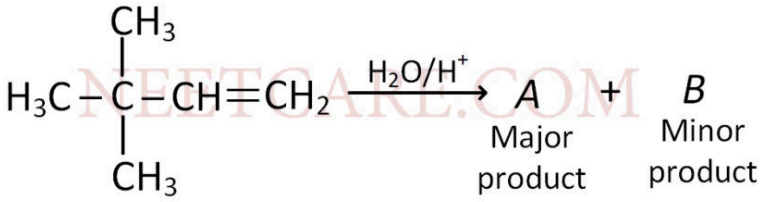
Open solution

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Q49. AIPMT 2012 Prelims

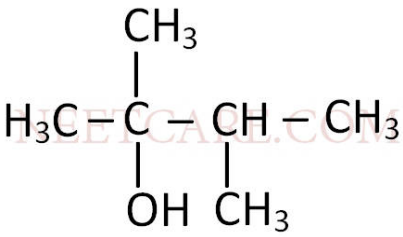
Open solution

In the following reaction

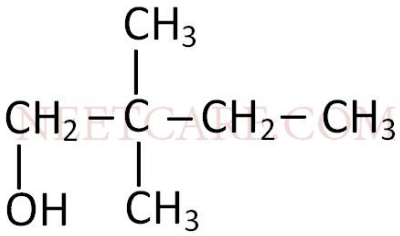


The major product is

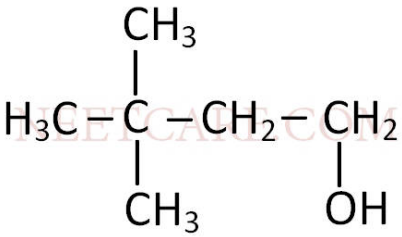
(A)



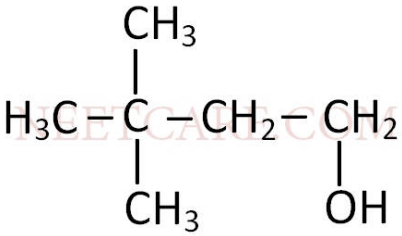
(B)



(C)



(D)



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Q50. AIPMT 2010 Prelims

Liquid hydrocarbons can be converted to a mixture of gaseous hydrocarbons by

(A) oxidation
(B) cracking
(C) distillation under reduced pressure
(D) hydrolysis

Open solution

Q51. AIPMT 2010 Prelims

The reaction of toluene with Cl_2 in presence of $FeCl_3$ gives X and reaction in presence of light gives Y. Thus, X and Y are

(A) X = Benzal chloride, Y = o-chlorotoluene
(B) X = m-chloroluene, Y = p-chlorotoluene
(C) X = o- and p-chlorotoluene, Y = Trichloromethyl benzene
(D) X = Benzyl chloride, Y = m-chlorotoluene

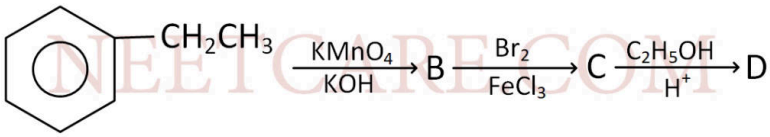
Open solution

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Q52. AIPMT 2010 Prelims

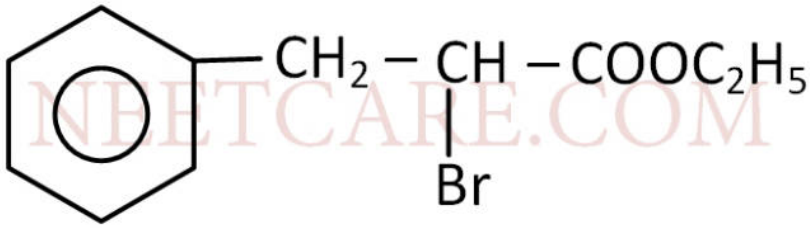
Open solution

In a set of reactions, ethylbenzene yielded a product D.

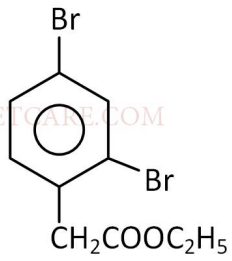


D would be

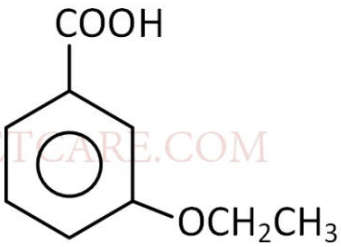
(A)



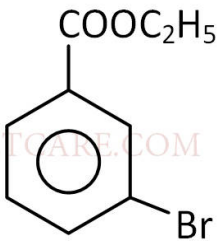
(B)



(C)



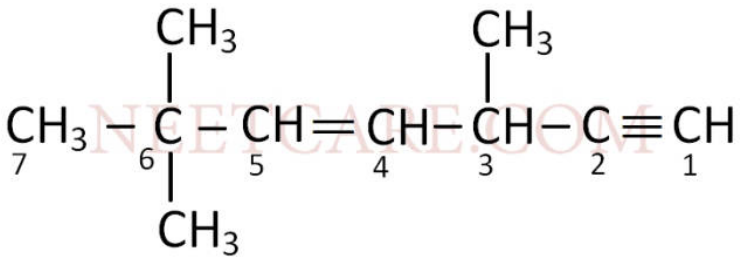
(D)



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Q53. AIPMT 2009

The state of hybridisation of C_2, C_3, C_5 and C_6 of the hydrocarbon,



is in the following sequence

(A) sp^3, sp^2, sp^2 and sp

(B) sp, sp^2, sp^2 and sp^3

(C) sp, sp^2, sp^3 and sp^2

(D) sp, sp^3, sp^2 and sp^3

Open solution

Q54. AIPMT 2009

Which of the following compounds will exhibit cis-trans (geometrical) isomerism?

(A) Butanol (B) 2-Butyne (C) 2-Butenol (D) 2-Butene

Open solution

Q55. AIPMT 2009

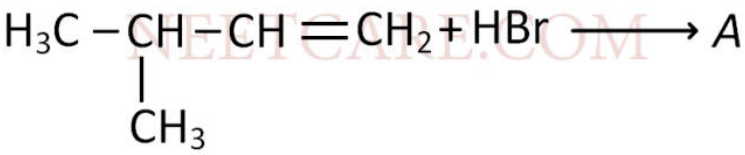
Benzene reacts with CH_3Cl in the presence of anhydrons $AlCl_3$ to form

(A) chlorobenzene (B) benzyl chloride (C) xylene (D) toluene.

Open solution

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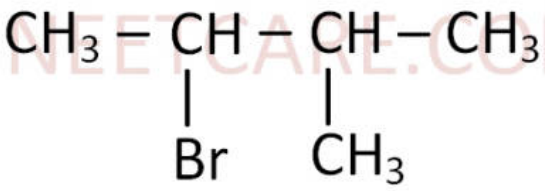
Q56. AIPMT 2008



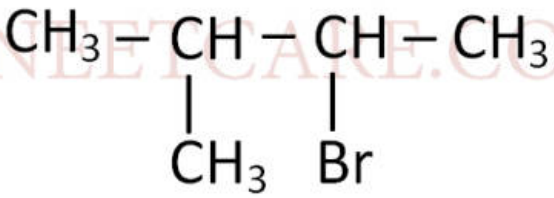
Open solution

A (predominantly) is

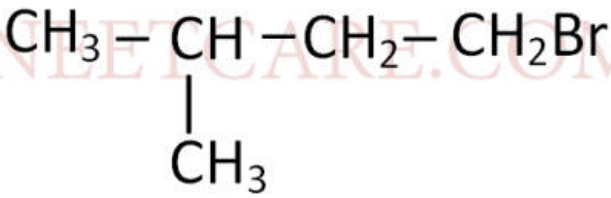
(A)



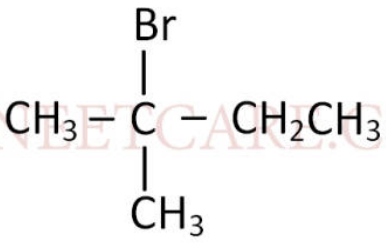
(B)



(C)



(D)



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Q57. AIPMT 2008

Open solution

In the hydrocarbon,



The state of hybridization of carbons 1, 3 and 5 are in the following sequence

(A) sp, sp^2, sp^3 (B) sp^3, sp^2, sp (C) sp^2, sp, sp^3 (D) sp, sp^3, sp^2

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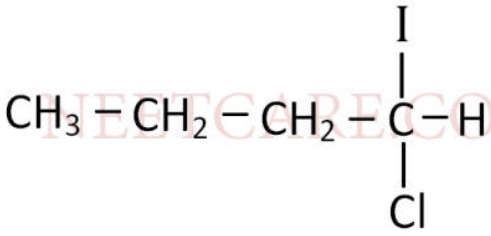
Q58. AIPMT 2007

Predict the product C obtained in the following reaction of 1-butyne.

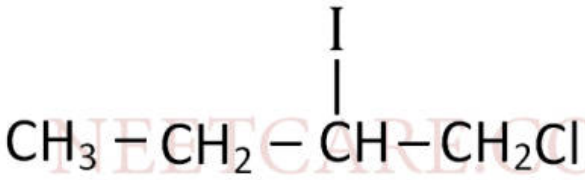


Open solution

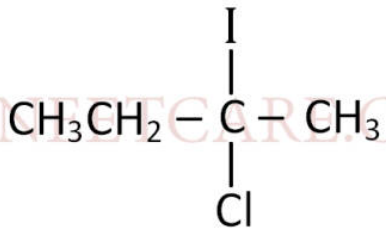
(A)



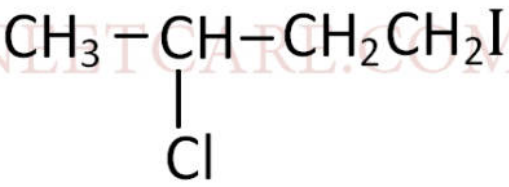
(B)



(C)



(D)



Q59. AIPMT 2007

Which of the compound with molecular formula C_5H_{10} yields acetone on ozonolysis?

(A) 3-Methyl- 1 -butene
(B) Cyclopentane
(C) 2-Methyl-1-butene
(D) 2-Methyl-2-butene

Open solution

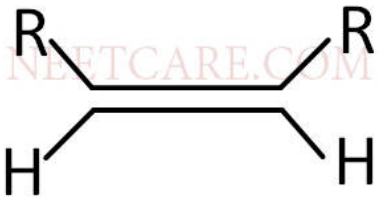
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Q60. AIPMT 2005

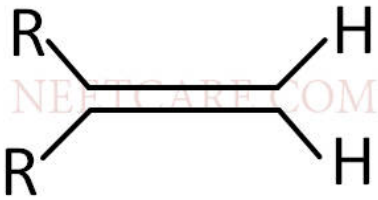
Open solution

Which one of the following alkenes will react faster with H_2 under catalytic hydrogenation conditions ?

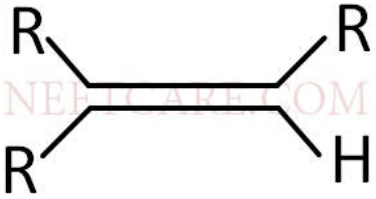
(A)



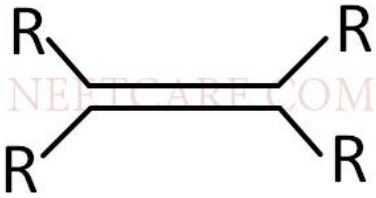
(B)



(C)



(D)



Q61. AIPMT 2005

Open solution

Products of the following reaction :

$$CH_3C\equiv CCH_2CH_3 \xrightarrow[\text{(ii) hydrolysis}]{\text{(i) } O_3} \text{are}$$

(A) $CH_3COOH + CO_2$

(B) $CH_3COOH + HOOC.CH_2CH_3$

(C) $CH_3CHO + CH_3CH_2CHO$

(D) $CH_3COOH + CH_3COCH_3$

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Q62. AIPMT 2004

Using anhydrous $AlCl_3$ as catalyst, which one of the following reactions produces ethylbenzene (PhEt)?

Open solution

(A) $H_3C - CH_2OH + C_6H_6$

(B) $CH_3 - CH = CH_2 + C_6H_6$

(C) $H_2C = CH_2 + C_6H_6$

(D) $H_3C - CH_3 + C_6H_6$

Q63. AIPMT 2004

Reaction of HBr with propene in the presence of peroxide gives

Open solution

(A) isopropyl bromide

(B) 3-bromopropane

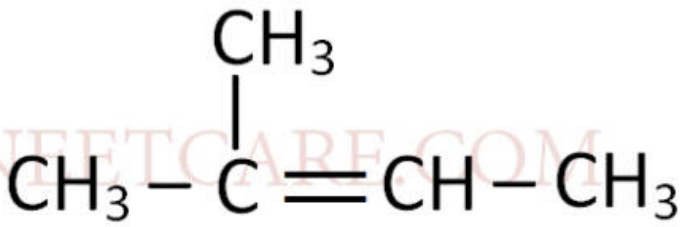
(C) allyl bromide

(D) n-propyl bromide

Q64. AIPMT 2003

The compound

Open solution



on reaction with $NaIO_4$ in the presence of $KMnO_4$ gives

(A) CH_3COCH_3

(B) $CH_3COCH_3 + CH_3COOH$

(C) $CH_3COCH_3 + CH_3CHO$

(D) $CH_3CHO + CO_2$

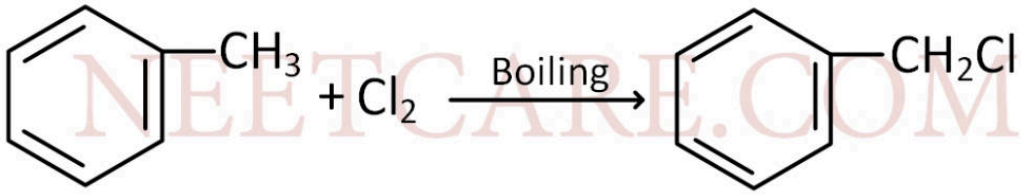
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Q65. AIPMT 2003

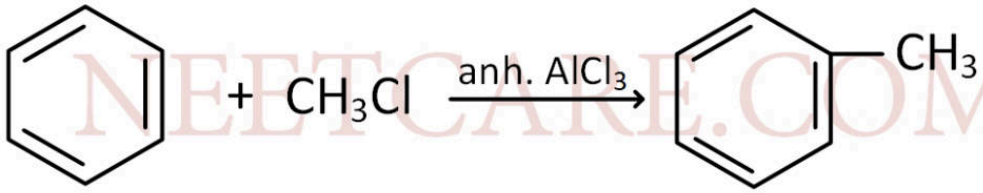
Open solution

Which one of the following is a free-radical substitution reaction ?

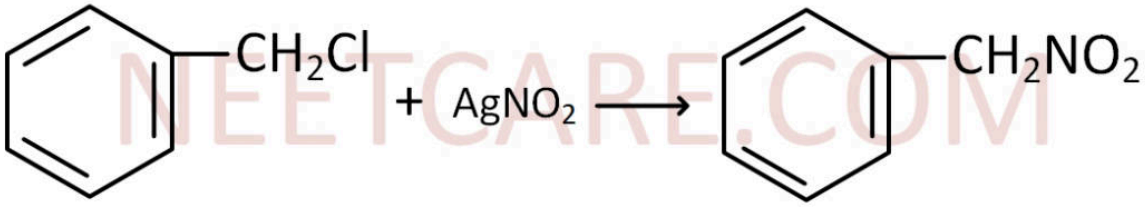
(A)



(B)



(C)



(D)



Q66. AIPMT 2003

Open solution

The correct order of reactivity towards the electrophilic substitution of the compounds aniline (I), benzene (II) and nitrobenzene (III) is

(A) III > II > I (B) II > III > I (C) I < II > III (D) I > II > III

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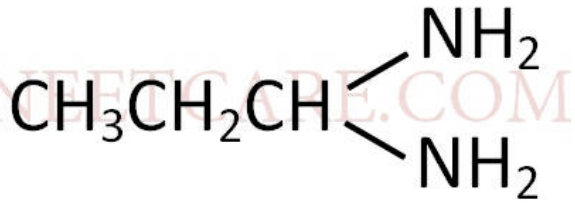
Q67. AIPMT 2002

When $CH_3CH_2CHCl_2$ is treated with $NaNH_2$, the product formed is

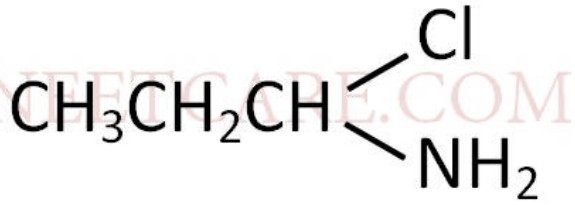
(A) $CH_3 - CH = CH_2$

(B) $CH_3 - C \equiv CH$

(C)



(D)



Open solution

Q68. AIPMT 2001

In preparation of alkene from alcohol using Al_2O_3 which is the effective factor ?

(A) Porosity of Al_2O_3

(B) Temperature

(C) Concentration

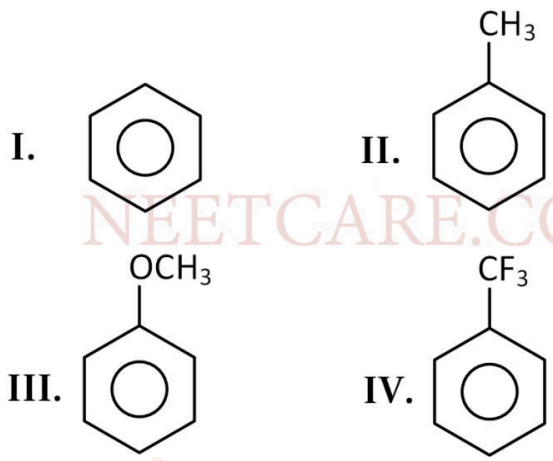
(D) Surface area of Al_2O_3

Open solution

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Q69. AIPMT 2000

Increasing order of electrophilic substitution for following compounds



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Open solution

(A) IV < I < II < III
(B) III < II < I < IV
(C) I < IV < III < II
(D) II < III < I < IV

Q70. AIPMT 2000

In Friedel-Crafts reaction , toluene can be prepared by

(A) $C_6H_6 + CH_3Cl$ (B) $C_6H_5Cl + CH_4$ (C) $C_6H_6 + CH_2Cl_2$ (D) $C_6H_6 + CH_3COCl$

Open solution

Q71. AIPMT 2000

Which reagent converts propene to 1-propanol ?

(A) H_2O, H_2SO_4
(B) B_2H_6, H_2O_2, OH^-
(C) $Hg(OAc)_2, NaBH_4/H_2O$
(D) Aq. KOH

Open solution

Q72. AIPMT 2000

Which is maximum stable?

(A) 1-Butene
(B) cis-2-Butene
(C) trans-2-Butene
(D) All have same stability.

Open solution

Q73. AIPMT 2000

2-Butene shows geometrical isomerism due to

(A) restricted rotation about double bond
(B) free rotation about double bond
(C) free rotation about single bond
(D) chiral carbon

Open solution

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Q74. AIPMT 2000

Dihedral angle in staggered form of ethane is

(A) 0° (B) 12° (C) 60° (D) 180°

Open solution

Q75. AIPMT 1999

When acetylene is passed through dil. H_2SO_4 in presence of $HgSO_4$, the compound formed is

(A) ether (B) ketone (C) acetic acid (D) acetaldehyde

Open solution

Q76. AIPMT 1999

In Friedel-Craft's alkylation, besides $AlCl_3$ the other reactants are

(A) $C_6H_6 + NH_2$ (B) $C_6H_6 + CH_4$ (C) $C_6H_6 + CH_3Cl$ (D) $C_6H_6 + CH_3COCl$

Open solution

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Q77. AIPMT 1999

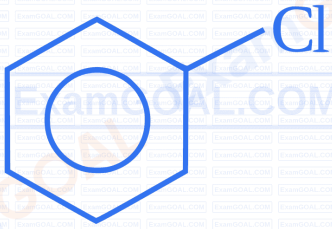
Which of the following compounds will be more easily attacked by an electrophile?

Open solution

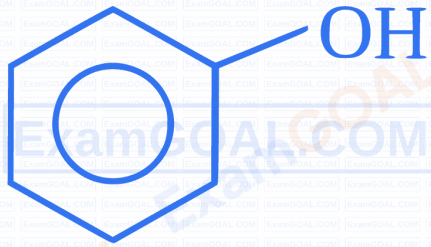
(A)



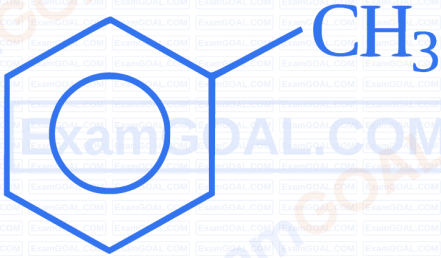
(B)



(C)



(D)



Q78. AIPMT 1998

Which one of these, is not compatible with arenes?

(A) Greater stability

(B) Delocalisation of π-electrons

(C) Electrophilic additions

(D) Resonance

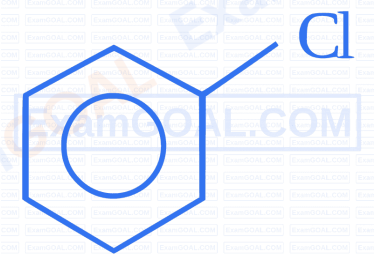
Open solution

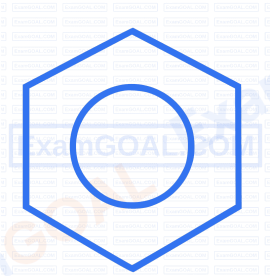
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Q79. AIPMT 1998

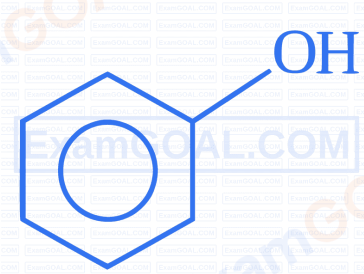
Open solution

Which one of the following compounds will be most easily attacked by an electrophile?

(A) 

(B) 

(C) 

(D) 

Q80. AIPMT 1997

Open solution

Which of the following reaction is expected to readily give a hydrocarbon product in good yields?

(A) $CH_3CH_3 \xrightarrow[h\nu]{Cl_2}$

(B) $(CH_3)_2CHCl \xrightarrow{C_2H_5OH}$

(C) $RCOOK \xrightarrow[\text{Oxidation}]{\text{Electrolysis}}$

(D) $RCOOAg \xrightarrow{I_2}$

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Q81. AIPMT 1997

In a reaction, $CH_2{=}CH_2 \xrightarrow[\text{acid}]{\text{Hypochlorous}} M \xrightarrow{R} \begin{matrix} CH_2OH \\ | \\ CH_2OH \end{matrix}$

where *M* = Molecule and *R* = Reagent. *M* and *R* are

(A) CH_3CH_2OH and HCl
(B) $CH_2{=}CH_2$ and heat
(C) CH_3CH_2Cl and NaOH
(D) $CH_2Cl{-}CH_2OH$ and aq. $NaHCO_3$.

Open solution

Q82. AIPMT 1997

The cylindrical shape of an alkyne is due to

(A) two sigma C – C and one π C – C bonds
(B) one sigma C – C and two π C – C bonds
(C) three sigma C – C bonds
(D) three π C – C bonds.

Open solution

Q83. AIPMT 1997

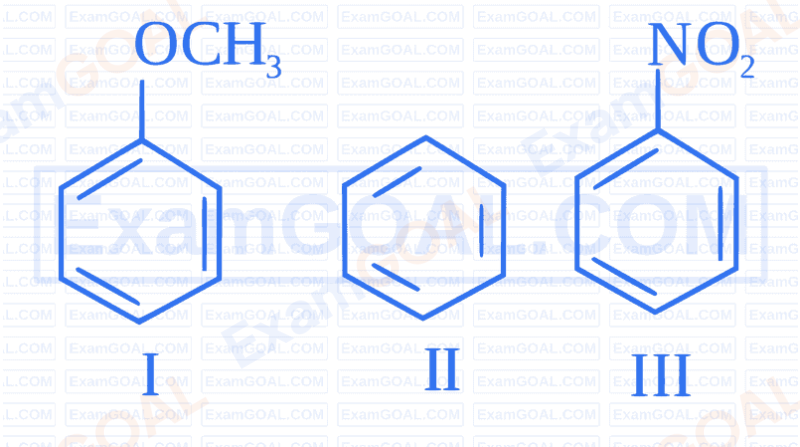
In the commercial gasolines, the type of hydrocarbons which are more desirable is

(A) linear unsaturated hydrocarbon
(B) toluene
(C) branched hydrocarbon
(D) straight-chain hydrocarbon.

Open solution

Q84. AIPMT 1997

Among the following compounds (I-III) the correct reaction with electrophile is



(A) $I > II > III$
(B) $I = II > III$
(C) $II > III > I$
(D) $III < I < II$

Open solution

Q85. AIPMT 1997

The most stable conformation of *n*-butane is

(A) gauche
(B) staggered
(C) skew boat
(D) eclipsed.

Open solution

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Q86. AIPMT 1996

Which of the following is used as an antiknocking material?

(A) Glyoxal (B) Freon (C) T.E.L. (D) Ethyl alcohol

Open solution

Q87. AIPMT 1996

Electrophile in the case of chlorination of benzene in the presence of $FeCl_3$ is

(A) Cl (B) $FeCl_3$ (C) Cl^+ (D) Cl^-

Open solution

Q88. AIPMT 1996

The reaction,

$CH_2{=}CH{-}CH_3 + HBr \rightarrow CH_3CHBr{-}CH_3$ is

(A) electrophilic substitution (B) free radical addition

(C) nucleophilic addition (D) electrophilic addition.

Open solution

Q89. AIPMT 1996

Which of the following has zero dipole moment?

(A) 1-Butene (B) 2-Methyl-1-propene

(C) *cis*-2-Butene (D) *trans*-2-Butene

Open solution

Q90. AIPMT 1995

The alkene $R{-}CH{=}CH_2$ reacts readily with B_2H_6 and the product on oxidation with alkaline hydrogen peroxides produces

(A) $R-\underset{\displaystyle CH_3}{\overset{\displaystyle |}{C}}=O$

(B) $R-\underset{\displaystyle OH}{\overset{\displaystyle |}{CH}}-\underset{\displaystyle OH}{\overset{\displaystyle |}{CH_2}}$

(C) $R{-}CH_2{-}CHO$

(D) $R{-}CH_2{-}CH_2{-}OH$

Open solution

Q91. AIPMT 1994

One of the following which does not observe the anti-Markownikoff's addition of HBr, is

(A) pent-2-ene (B) propene (C) but-2-ene (D) but-1-ene

Open solution

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Q92. AIPMT 1994

The reactive species in the nitration of benzene is

(A) NO_3 (B) HNO_3 (C) NO_2^+ (D) NO_2^-

Open solution

Q93. AIPMT 1993

The restricted rotation about carbon carbon double bond in 2-butene is due to

(A) overlap of one *s* and sp^2-hybridized orbitals

(B) overlap of two sp^2-hybridized orbitals

(C) overlap of one *p* and one sp^2-hybridized orbitals

(D) sideways overlap of two *p*-orbitals.

Open solution

Q94. AIPMT 1993

$R-CH_2-CCl_2-R \xrightarrow{\text{Reagent}} R-C\equiv C-R$

The reagent is

(A) Na (B) HCl in H_2O (C) KOH in C_2H_5OH (D) Zn in alcohol.

Open solution

Q95. AIPMT 1993

Reduction of 2-butyne with sodium in liquid ammonia gives predominantly

(A) *cis*-2-butene (B) no reaction (C) *trans*-2-butene (D) *n*-butane.

Open solution

Q96. AIPMT 1993

A compound is treated with $NaNH_2$ to give sodium salt. Identify the compound.

(A) C_2H_2 (B) C_6H_6 (C) C_2H_6 (D) C_2H_4

Open solution

Q97. AIPMT 1993

Reactivity of hydrogen atoms attached to different carbon atoms in alkanes has the order

(A) tertiary $>$ primary $>$ secondary (B) primary $>$ secondary $>$ tertiary

(C) both (1) and (2) (D) tertiary $>$ secondary $>$ primary.

Open solution

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Q98. AIPMT 1993

Which is the correct symbol relating the two Kekule structures of benzene?

(A) $\rightleftharpoons$

(B) $\longrightarrow$

(C) $\equiv$

(D) $\longleftrightarrow$

Open solution

Q99. AIPMT 1993

Benzene reacts with *n*-propyl chloride in the presence of anhydrous $AlCl_3$ to give

(A) 3-propyl-1-chlorobenzene

(B) *n*-propylbenzene

(C) no reaction

(D) isopropylbenzene.

Open solution

Q100. AIPMT 1993

When hydrochloric acid gas is treated with propene in presence of benzoyl peroxide, it gives

(A) 2-chloropropane

(B) allyl chloride

(C) no reaction

(D) n-propyl chloride

Open solution

Q101. AIPMT 1992

Among the following compounds the one that is most reactive towards electrophilic nitration is

(A) benzoic acid

(B) nitrobenzene

(C) toluene

(D) benzene

Open solution

Q102. AIPMT 1992

Select the true statement about benzene amongst the following

(A) because of unsaturation benzene easily undergoes addition

(B) there are two types of C – C bonds in benzene molecule

(C) there is cyclic delocalisation of π-electrons in benzene

(D) monosubstitution of benzene gives three isomeric products.

Open solution

Q103. AIPMT 1989

Cyclic hydrocarbon '*A*' has all the carbon and hydrogen atoms in a single plane. All the carbon-carbon bonds have the same length, less than 1.54 Å, but more than 1.34 Å. The bond angle will be

(A) $109^\circ 28'$

(B) 100°

(C) 180°

(D) 120°

Open solution

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Q104. AIPMT 1989

Open solution

Acetylenic hydrogens are acidic because

(A) sigma electron density of C – H bond in acetylene is nearer to carbon, which has 50% *s*-character

(B) acetylene has only open hydrogen in each carbon

(C) acetylene contains least number of hydrogens among the possible hydrocarbons having two carbons

(D) Acetylene belongs to the class of alkynes with molecular formula, C_nH_{2n-2}.

Q105. AIPMT 1989

Open solution

Which is the most suitable reagent among the following to distinguish compound (3) from rest of the compounds?

(1) $CH_3-C\equiv C-CH_3$

(2) $CH_3-CH_2-CH_2-CH_3$

(3) $CH_3-CH_2C\equiv CH$

(4) $CH_3-CH=CH_2$

(A) Bromine in carbon tetrachloride

(B) Bromine in acetic acid

(C) Alk. $KMnO_4$

(D) Ammoniacal silver nitrate.

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Answer Key

1. **A**	2. **B**	3. **B**	4. **C**	5. **B**
6. **C**	7. **C**	8. **B**	9. **B**	10. **D**
11. **A**	12. **C**	13. **D**	14. **A**	15. **A**
16. **D**	17. **B**	18. **C**	19. **B**	20. **D**
21. **D**	22. **A**	23. **D**	24. **A**	25. **C**
26. **C**	27. **D**	28. **A**	29. **D**	30. **C**
31. **C**	32. **A**	33. **D**	34. **D**	35. **B**
36. **C**	37. **A**	38. **C**	39. **A**	40. **C**
41. **D**	42. **C**	43. **A**	44. **B**	45. **C**
46. **A**	47. **A**	48. **C**	49. **A**	50. **B**
51. **C**	52. **D**	53. **D**	54. **D**	55. **D**
56. **D**	57. **D**	58. **C**	59. **D**	60. **A**
61. **B**	62. **C**	63. **D**	64. **B**	65. **A**
66. **D**	67. **B**	68. **B**	69. **A**	70. **A**
71. **B**	72. **C**	73. **A**	74. **C**	75. **D**
76. **C**	77. **C**	78. **C**	79. **D**	80. **C**
81. **D**	82. **B**	83. **C**	84. **A**	85. **B**
86. **C**	87. **C**	88. **D**	89. **D**	90. **D**
91. **C**	92. **C**	93. **D**	94. **C**	95. **C**
96. **A**	97. **D**	98. **D**	99. **D**	100. **A**
101. **C**	102. **C**	103. **D**	104. **A**	105. **D**

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